

Supplementary material

Incremental Value of Daily Workload and Wellness Monitoring Beyond a Personalised Readiness-History Baseline for Next-Calendar-Day Readiness in Elite Women’s Football

Table S1: Model-specific full-pipeline tuning in source-team 2020 data. Candidate penalties were $\alpha = 0, 0.0001, 0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000, 100000$ and were evaluated in three expanding chronological folds. Selection used weighted validation MAE after reset sequential residual adaptation.

Source team	Model	Selected α	Weighted validation MAE	Validation pairs
Team A	P0	1000	0.797	1,792
Team A	P1	1000	0.786	1,792
Team A	P2	1000	0.783	1,792
Team B	P0	1000	0.566	1,064
Team B	P1	1000	0.576	1,064
Team B	P2	1000	0.577	1,064

Table S2: Residual-window and reduced-monitoring sensitivity analyses. Positive values indicate higher MAE for the candidate model than for the personalised readiness-history baseline.

Candidate model	Evaluation setting	Residual window	Test pairs	Candidate- $P0$ MAE (95% CI)
P1 full monitoring	Within Team A: 2020→2021	14	5,881	+0.010 (-0.003 to +0.027)
P1 full monitoring	Within Team A: 2020→2021	28	5,881	+0.007 (-0.005 to +0.023)
P1 full monitoring	Within Team A: 2020→2021	56	5,881	+0.004 (-0.007 to +0.020)
P1 full monitoring	Within Team B: 2020→2021	14	2,483	+0.021 (-0.003 to +0.049)
P1 full monitoring	Within Team B: 2020→2021	28	2,483	+0.018 (-0.005 to +0.047)
P1 full monitoring	Within Team B: 2020→2021	56	2,483	+0.014 (-0.011 to +0.043)
P1 full monitoring	Team A 2020→Team B 2021	14	2,483	+0.028 (+0.001 to +0.060)
P1 full monitoring	Team A 2020→Team B 2021	28	2,483	+0.022 (-0.005 to +0.053)
P1 full monitoring	Team A 2020→Team B 2021	56	2,483	+0.017 (-0.012 to +0.052)
P1 full monitoring	Team B 2020→Team A 2021	14	5,881	+0.004 (-0.007 to +0.017)
P1 full monitoring	Team B 2020→Team A 2021	28	5,881	+0.004 (-0.007 to +0.016)
P1 full monitoring	Team B 2020→Team A 2021	56	5,881	-0.000 (-0.009 to +0.010)
P2 reduced monitoring	Within Team A: 2020→2021	28	5,881	+0.011 (-0.002 to +0.027)
P2 reduced monitoring	Within Team B: 2020→2021	28	2,483	+0.017 (-0.006 to +0.044)
P2 reduced monitoring	Team A 2020→Team B 2021	28	2,483	+0.023 (-0.001 to +0.052)
P2 reduced monitoring	Team B 2020→Team A 2021	28	5,881	+0.002 (-0.007 to +0.013)

Table S3: Performance and linear calibration diagnostics under the primary 28-observation residual-history window. The intercept and slope are from regressions of observed readiness on predicted readiness.

Evaluation setting	Model	α	MAE	RMSE	Bias	R^2	Intercept	Slope	Within ± 1 point
Within Team A: 2020→2021	<i>P0</i>	1000	0.717	1.031	0.012	0.568	0.472	0.928	75.9%
Within Team A: 2020→2021	<i>P1</i>	1000	0.724	1.031	0.032	0.568	0.504	0.920	76.0%
Within Team B: 2020→2021	<i>P0</i>	1000	0.619	0.965	0.028	0.486	0.624	0.908	81.1%
Within Team B: 2020→2021	<i>P1</i>	1000	0.637	0.956	0.031	0.496	0.545	0.919	80.9%
Team A 2020→Team B 2021	<i>P0</i>	1000	0.618	0.962	0.029	0.490	0.502	0.925	81.3%
Team A 2020→Team B 2021	<i>P1</i>	1000	0.640	0.969	0.058	0.482	0.525	0.918	81.1%
Team B 2020→Team A 2021	<i>P0</i>	1000	0.718	1.034	0.012	0.565	0.551	0.916	75.7%
Team B 2020→Team A 2021	<i>P1</i>	1000	0.722	1.031	0.017	0.567	0.531	0.918	76.0%

Table S4: Player-day eligibility flow from the complete calendar panel to the primary paired complete-case analytic cohort. All player-days with observed current- and next-day readiness already had the required 7-day load history because the daily-load calendar panel was continuous.

Step	Overall player-days	Overall players	Team A player-days	Team A players	Team B player-days	Team B players
Full calendar player-day panel	36,550	50	19,737	27	16,813	23
Observed next-day readiness	16,997	50	11,022	27	5,975	23
Observed current- and next-day readiness plus 7-day load history	14,631	50	9,722	27	4,909	23
Plus complete current-day readiness and wellness profile	14,582	50	9,693	27	4,889	23

Table S5: Missingness of current-day readiness/wellness variables and next-day readiness in the calendar player-day panel, by team and year. Values are percentages of player-days with missing values.

Team-year	Current-day readiness	Fatigue	Mood	Sleep duration	Sleep quality	Soreness	Stress	Next-day readiness
Team A, 2020	56.2	56.2	56.2	56.2	56.2	56.2	56.2	56.1
Team A, 2021	32.1	32.1	32.1	32.2	32.1	32.1	32.1	32.1
Team B, 2020	67.3	67.3	67.3	67.3	67.3	67.3	67.3	67.3
Team B, 2021	61.6	61.6	61.6	61.6	61.6	61.6	61.6	61.6

Table S6: Wellness-item-missingness sensitivity-model tuning in source-team 2020 data. The full model used source-training median imputation and wellness-specific missingness indicators.

Source team	Model	Selected α	Weighted validation MAE	Validation pairs
Team A	P0	1000	0.797	1,796
Team A	P1m	1000	0.787	1,796
Team B	P0	1000	0.565	1,065
Team B	P1m	1000	0.575	1,065

Table S7: Wellness-item-missingness sensitivity analysis conditional on observed current- and next-day readiness and 7-day load history. Positive values indicate higher MAE for the imputed full model (P1m) than for the personalised readiness-history baseline. This analysis does not address days on which the daily readiness questionnaire was not completed.

Evaluation setting	Test pairs	P0 MAE	P1m MAE	P1m–P0 MAE (95% CI)
Within Team A: 2020→2021	5,903	0.716	0.723	+0.008 (-0.004 to +0.025)
Within Team B: 2020→2021	2,497	0.617	0.636	+0.020 (-0.006 to +0.049)
Team A 2020→Team B 2021	2,497	0.616	0.639	+0.024 (-0.003 to +0.059)
Team B 2020→Team A 2021	5,903	0.716	0.720	+0.004 (-0.006 to +0.016)

Table S8: Evaluation eligibility and the scope of the wellness-item-missingness sensitivity analysis. Counts are retrospective because they condition on subsequently observed next-day readiness; they should not be interpreted as prospective deployment-availability rates.

Evaluation subset	Player-day pairs	Percentage of full calendar panel
Full calendar player-day panel	36,550	100.0%
Observed next-day readiness	16,997	46.5%
Observed current- and next-day readiness plus 7-day load history	14,631	40.0%
Primary paired complete-case cohort (plus complete current-day readiness/wellness profile)	14,582	39.9%
Additional pairs admitted by item-level wellness imputation	49	0.1%

Note: The 49 additional pairs equal 0.3% of the 14,631-pair conditional cohort. The primary P0–P1 comparison uses the same 14,582 report-complete player-days to avoid changing the evaluation set between models.

Table S9: Implementation and interpretation scope of the forecasting evaluation.

Element	Implemented approach	Interpretation boundary
Prediction target	Self-reported readiness at calendar day $t + 1$ on a 1–10 scale.	Not a performance, injury, or clinical outcome.
Information timing	Candidate inputs available by the end of calendar day t ; target aligned to the next calendar date for the same player.	The public data lack complete within-day timestamps; this is not a fully timestamp-verified real-time simulation.
Population-model develop-ment	Source-team 2020 data only; model-specific expanding chronological tuning; source-only pre-processing.	No 2021 target outcome is used to select penalties.
Sequential personalisation	Up to 28 earlier eligible target-period residuals for the same player; reset at target-period start.	Appropriate for ongoing monitoring; not a true new-player cold-start evaluation.
Cross-team assessment	Population parameters transfer between teams, followed by target-team residual updating when earlier outcomes accrue.	Transportability assessment under sequential adaptation, not zero-shot external validation.
Primary comparison	$P1$ minus $P0$ MAE on the same 14,582 report-complete player-days.	Isolates added panel information beyond identical current-readiness and residual-history information.
Uncertainty	2,000 percentile player-cluster bootstrap resamples.	Quantifies uncertainty across players within the target team, not across teams.
Missingness sensitivity	Wellness-item median imputation plus missingness indicators in 14,631 otherwise evaluable pairs.	Adds 49 pairs and does not model days with an absent daily readiness questionnaire.
Calendar representation	Numeric weekday and month terms in linear/ridge models.	May not capture nonlinear or cyclical calendar effects.
Prediction range	Continuous predictions; no rounding or truncation to 1–10.	A deployed system may require a pre-specified range-handling rule.

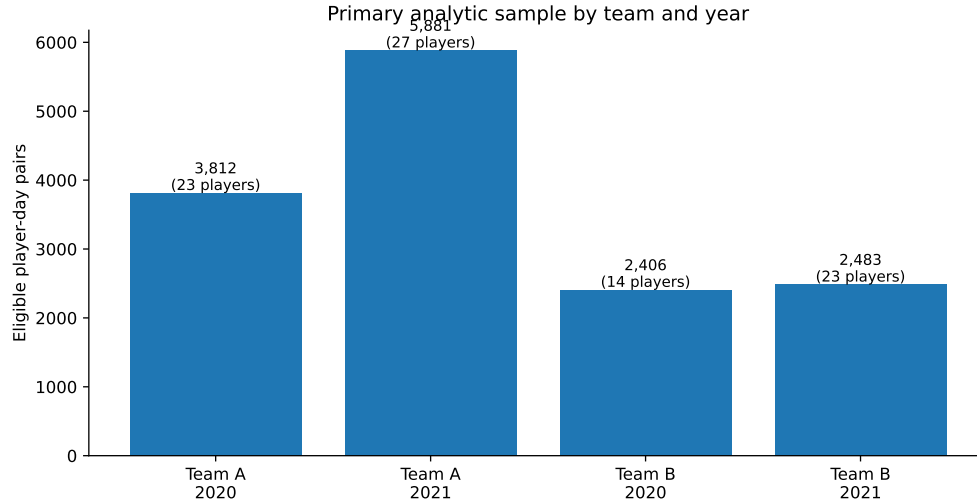


Figure S1: Primary analytic sample by team and year. Labels show the number of eligible player-day pairs and contributing players.

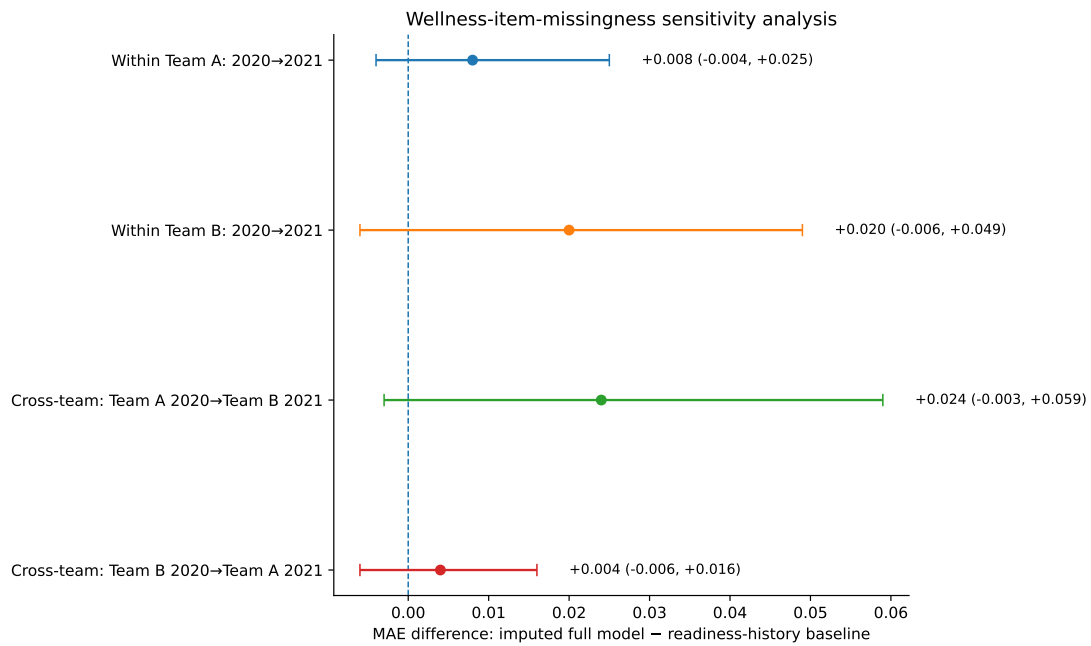


Figure S2: Wellness-item-missingness sensitivity analysis conditional on observed current- and next-day readiness. The full model used source-training median imputation and wellness-specific missingness indicators. Positive values indicate higher MAE than the personalised readiness-history baseline.

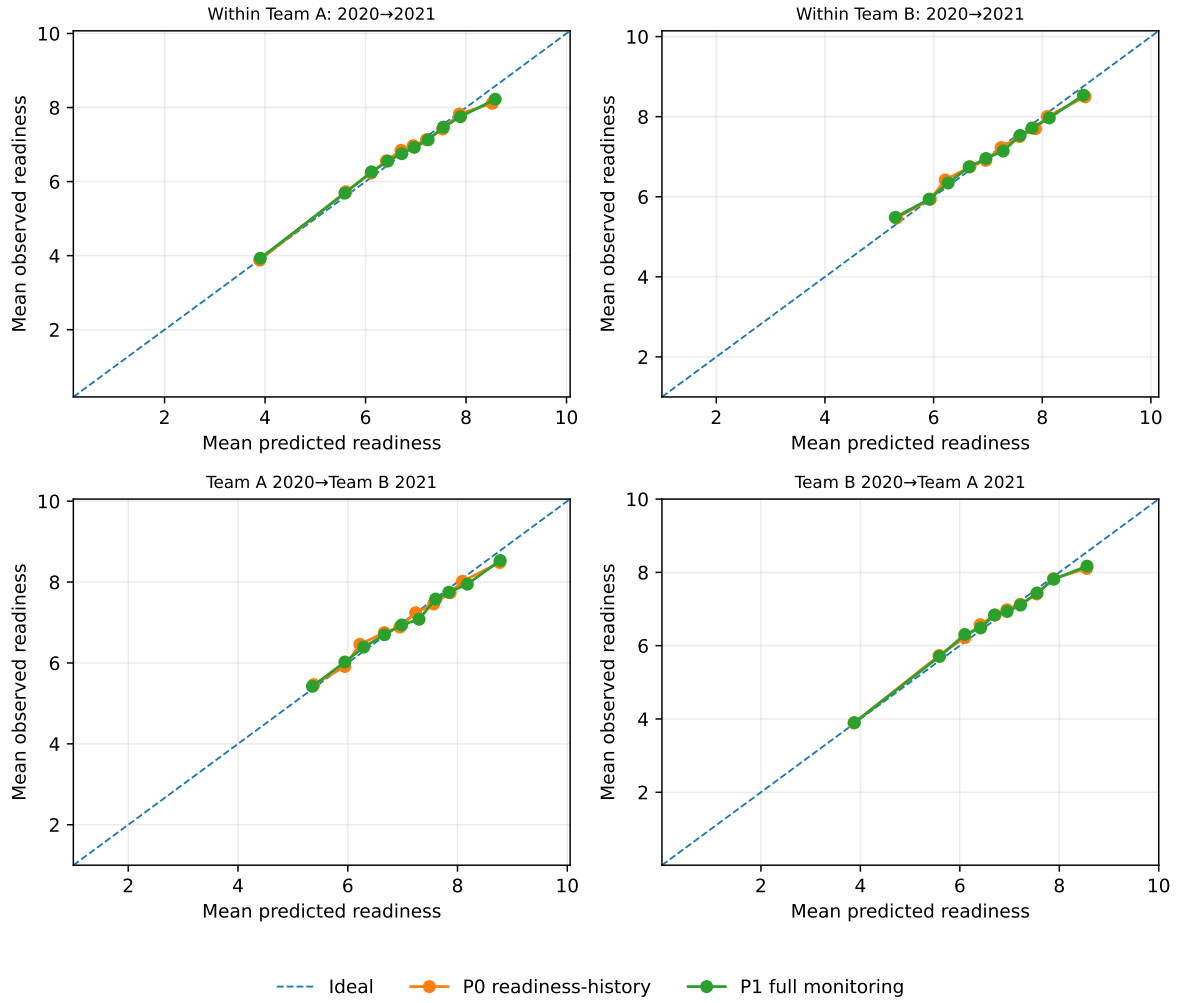


Figure S3: Decile-based calibration curves for $P0$ and $P1$ under the primary 28-observation sequential residual-history window. Points show mean observed readiness within prediction-rank bins; the dashed line indicates ideal agreement.

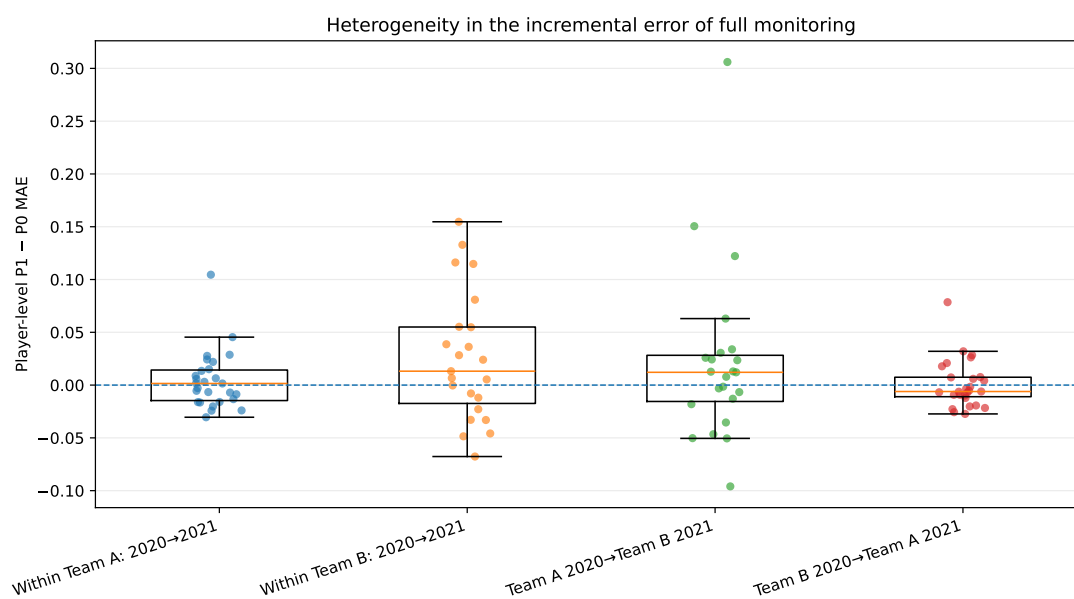


Figure S4: Player-level heterogeneity in full-monitoring incremental MAE. Each point is one player. Positive values indicate higher MAE for $P1$ than for $P0$. Boxplots summarize player-level distributions within each evaluation setting; this descriptive plot does not identify stable player-level treatment or causal effects.